

Mapping the Tool Landscape for Stakeholder Involvement in Participatory AI: Strengths, Gaps, and Future Directions

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Abstract

Stakeholder Involvement (SHI) during technology development is increasingly promoted in responsible AI (rAI) guidance. However, the lack of concrete, actionable tools to realise this is a well-known issue. To inform the future development of such SHI-supporting tools, this study presents an analysis of the existing tool landscape. We reviewed 216 rAI tools (covering seven meta-reviews, 2020–2024), revealing that only 18% provide actionable support for SHI. Mapping these tools to both the SHI process and the AI lifecycle, we found a strong focus on support when communicating with stakeholders, especially during early ideation stages. Tools supporting many other practical aspects of SHI were rare (e.g. recruiting stakeholders or storing insights) as were tools supporting reactions to gained insights and tools for participation during the testing stage. We point practitioners towards existing tool support and recommend concrete actions and immediate research agendas to address the exposed gaps, facilitating rAI-aligned SHI going forward.

CCS Concepts

• **Human-centered computing** → **Participatory design; Systems and tools for interaction design.**

Keywords

Participatory AI Development, Stakeholder Involvement, Responsible AI, Practitioner Tools

ACM Reference Format:

Emma Kallina and Jatinder Singh. 2025. Mapping the Tool Landscape for Stakeholder Involvement in Participatory AI: Strengths, Gaps, and Future Directions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719726>

1 Introduction

A surge of recent work promotes stakeholder involvement throughout the AI lifecycle as an important aspect of responsible AI (rAI)

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719726>

development [e.g. 2, 38, 41, 44, 60, 66]. *Stakeholders* in the software development context are described as an “[i]ndividual or organization [...] having a right, share, claim, or interest in a system or in its possession of characteristics that meet their needs and expectations” [section 3.44, ISO 15288, 36]. Often, key stakeholders are users or domain experts, however, in this work we specifically also include affected non-users or other groups at risk [47]. *Stakeholder Involvement* (SHI) entails engaging a system’s stakeholders at various points along a system’s lifecycle to explore their needs, values, and concerns in the application context to then address them in the system’s design [63].

Recent works associate SHI during AI development with various benefits such as a better understanding of the system’s socio-technical context [13, 39, 41, 73] and thus the improved detection of potential risks and harms [18, 41, 72]: involving affected and especially at-risk communities can shed light on context-specific details and existing inequalities that are likely to be exacerbated by AI systems [38, 41, 72]. This can alleviate the opacity of AI systems that currently impedes predictions on how they will operate once deployed; especially if deployed at scale [62]. Further, by shifting decision power during development towards the communities the system will affect, SHI can expand the values and mindsets on which such a system is based beyond those of its creators to include (or ideally be based on) the values of affected communities [10, 14, 38, 64]. For example, if public sector AI would be based more on citizens’ needs than the needs of government agencies (i.e. the systems’ customers), we might be able to move away from a focus on policing (e.g. fraud detection, predictive policing [12, 29, 68, 74]) and towards systems that help citizens to obtain better services. Considering these benefits, it is unsurprising that also various institutions include SHI in their rAI guidance documents, such as intergovernmental or multi-stakeholder organisations (e.g. UN [21, 22], OECD [50], WHO [53]; ACM [51, 52]) or civil society organisations (Amnesty International [33–35], Access Now [1, 49, 54], AlgorithmWatch [3, 4]). However, it is important to note that SHI does not only require specific expertise in SHI methods as well as significant resource-, staff-, and time-investments, but also rarely delivers a ‘silver bullet’ solution to rAI: different communities might have different needs that force difficult trade-offs [for detailed challenges, see 38]). Thus, SHI is a challenging process.

Despite the growing calls for SHI to advance rAI efforts, the lack of concrete, actionable support—i.e. *tools*—to realise this challenging task is a well-known issue [e.g. 45, 60, 61, 71]. Such tools (e.g.

decks of cards, canvases, or checklists) are designed to translate academic or best-practice guidance to real-life projects, facilitating their implementation in practice [30, 48]. Here, *actionable* implies that they go beyond mentioning or recommending a specific action, but provide more concrete advice or steps on how to actually take this action. This study **analyses existing rAI tools to identify (1) how many of those focus on supporting responsible SHI practices during AI development, (2) which aspects of SHI are already well supported, and (3) current gaps in such tool support.** Our analysis provides essential insights to direct practitioners towards existing tools as well as to improve such tool support going forward. More specifically, we reviewed 216 rAI tools from seven tool meta-reviews (2020 to 2024) to identify which actions and aspects of SHI and the AI lifecycle are currently well-supported and which lack support. This provides the following contributions:

- (1) **Highlighting areas that require urgent research attention**, as is essential to inform more actionable support going forward. More specifically, we find that:
 - (a) *Only a small fraction of rAI tools provide actionable support* for rAI development, emphasising the disconnect to growing calls for SHI in rAI literature.
 - (b) *Many actions of SHI require more support* as most tools focus on the SHI session alone (vs preparing or reacting to it) – a clear opportunity for the HCI community to further explore these bottlenecks and to co-design tools with practitioners.
 - (c) *Tools that support reactions to gathered insights are rare*, i.e. tools that ensure that organisations actually react to gathered insights. Designing such tools is a matter of urgency as they are required to move towards SHI practices in line with rAI efforts.
 - (d) *Show the need of practitioner support beyond increased awareness* (i.e. the focus of tools) such as more incentives to engage in responsible SHI practices.
- (2) **Pointing practitioners to currently available tool support** to immediately facilitate more responsible SHI.
 - (a) Early AI lifecycle stages are well-supported, especially when exploring the needs and values of stakeholders.
- (3) **Pointing towards concrete actions and research questions** for the HCI and rAI communities that are direct reactions to the findings of this study, further facilitating rAI-aligned SHI going forward.

2 Background: rAI Literature on SHI and Tools

2.1 SHI in rAI Development

The rAI literature investigates and encourages SHI during AI development as a means to prevent harmful or discriminatory impacts of the resulting systems [see e.g. 11, 16, 19, 38, 39], a reaction to the growing evidence that AI can compromise fundamental rights with an increased risk to marginalised communities [e.g. 18, 46, 69]. A surge of recent work associates SHI with various benefits around a more equal distribution of the harms and benefits of AI systems [e.g. 10, 13, 18, 38, 39, 72]. This led a wide range of rAI guidance documents to encourage SHI, including the ones listed above. However, to actually harness these benefits, (i) a wide range of stakeholders must be involved (i.e. beyond revenue-critical stakeholders

and including at-risk groups) and (ii) these stakeholders have to be empowered to fundamentally shape key aspects of the system, including its objective [10, 14, 38]. Such SHI goes beyond more commercially-oriented SHI such as usability studies or customer meetings [10, 14, 16]. *Responsible SHI practices*, i.e. SHI that contributes to rAI efforts and tools supporting such are the focus of this study.

Besides these increasing calls for SHI across the rAI literature and guidance, insights into the practice suggest that currently responsible SHI is rarely enacted: AI development in research contexts rarely empowers involved stakeholders to shape key aspects of the system [14, 17, 19] and public engagement during commercial AI projects is scarce and under-resourced, even when individual employees are promoting this approach [26]. Taken together, there seems to be a **substantial gap between the recommendations for responsible SHI and its implementation in the AI practice.** We hope that this study helps to overcome this gap by pointing towards existing support mechanisms as well as by identifying opportunities for better practitioner support.

2.2 Tools Supporting the Implementation of rAI Development Practices

rAI tools are designed to guide and support practitioners during the implementation of rAI development and deployment practices and can take various forms, e.g. decks of cards, canvases, checklists, technical metrics, or entire toolkits [30, 48, 56]. Previous work has reviewed such general (i.e. not SHI-specific) rAI tools, e.g. by mapping them to the rAI principles such as fairness or values they aim to advance [48, 56]. These studies found a strong focus on rAI principles that can be addressed with technical measures such as explainability. Tools for more qualitative, perception-based aspects (thus more relevant for SHI) were not only less established, but they were also less concrete and actionable [48, 56]. Other rAI tool reviews analysed the tools' intended audiences, i.e. the roles that would apply these tools [6, 70]. While SHI was not the focus of these analyses, Ayling and Chapman [6] noted that there is "little participation [...] by certain stakeholder groups (Voiceless, Vested Interests and Users) who are not included in the process of applying the tools or interacting with the outputs" – suggesting that SHI with affected communities is currently not well supported. Further, a study by [9] analysed several studies that evaluated rAI tools and found that these are often limited to assessing the tools' usability while other aspects are not addressed, e.g. whether the involved stakeholders are actually empowered to shape the system. Relating these findings to the literature around SHI, it seems that **SHI—particularly responsible SHI—remains underserved by existing rAI tools.** This work is designed to better analyse this gap to inform future practitioner support.

3 Methods

The following section explains the methods underlying our analysis of the current landscape of tools designed to support practitioners during SHI for rAI efforts.

3.1 Selection of Tools

To compile a broad list of SHI-specific tools available to AI practitioners, we first created a list of tools specifically focused on supporting rAI practices. To do so, we reviewed the 13 meta-reviews of rAI tools and toolkits discussed in Hollanek [30], i.e. [5, 6, 15, 20, 25, 27, 37, 40, 48, 56–59, 71]. Due to our focus on tools—i.e. the concrete, actionable implementation of rAI aspects—we removed six reviews that focused on high-level rAI principles or guidance only. We further excluded two reviews focusing on technical measures (e.g., code libraries) as they are unlikely to facilitate SHI [i.e. 40, 58]. Complementing the remaining six meta-reviews, we performed a keyword search on Google Scholar to identify further tool reviews, utilising a combination of keywords around (i) responsible or ethical AI, (ii) participatory design or stakeholder/community involvement, and (iii) tools or toolkits. This produced one additional tool meta-review, i.e. a review of 63 ethics-focused design tools supporting software development [13]. Although this review was not explicitly focused on AI systems, the included tools were highly relevant for our analysis as they emphasised responsible SHI practices. Thus, we decided to include this review, further motivated by the fact that also some of the AI-specific reviews included some tools that were not explicitly focused on AI (e.g. Wong et al. [71] included Microsoft’s Community Jury [7]). By retrieving the tools reviewed in at least one of the seven meta-reviews illustrated in Table 1, we generated a list of 235 tools, many of which were named in multiple reviews. Of these 235 tools, we excluded 19 tools due to following reasons: (i) we could not actually find them even if using different search engines ($n=11$), (ii) they were not actually tools but instead communities or organisations in the rAI space ($n=6$), (iii) they had a paywall ($n=1$), or (iv) were not available in English ($n=1$). This left us with 216 tools for our further analysis. A Table listing all tools, their sources, and other details is included in the supplementary materials.

3.2 Mapping & Analysis

To identify which of the 216 rAI tools actually supported responsible SHI, we reviewed each tool and assigned it to one of following four, mutually exclusive categories:

Category 1: Provides **explicit and actionable support** for at least one aspect of SHI.

Category 2: Explicitly **promotes SHI but it does not provide actionable support**. This category differs from Category 1 by merely mentioning SHI/actions related to SHI (e.g. mentioning that affected at-risk communities should be involved) without providing more concrete guidance on how to realise this in practice.

Category 3: **Potentially dis-incentivises SHI** by encouraging the consideration of suspected stakeholder needs (e.g. via personas) without explicitly clarifying that such considerations have to include and empower the stakeholders in focus.

Category 4: Tools that **do not mention SHI**, i.e. tools that addressed aspects of rAI without any stakeholder input, often through technical methods, e.g. aimed at improving *explainability* through technical XAI methods.

The next step was designed to understand which aspects of the SHI process are currently well supported vs where support is insufficient. To gather systematic insights on two crucial aspects, we mapped the tools from Category 1 to two different processes whereby a tool

could support more than one stage per process. The first process was the **AI lifecycle** after Ayling and Chapman [6]. Since the different stages of the AI lifecycle require different SHI methods (e.g. more open, exploratory methods at early stages and more specific, prototype-based methods towards later stages, [31]), this mapping allowed us to identify which stages are currently well-supported and which are not yet considered. The second process is the **responsible SHI process** after Kallina and Singh [38] (Fig. 2). Independent of the AI lifecycle stage, an instance of SHI follows a specific process from planning the SHI session over conducting it and reacting to the gathered insights [38]. By examining these stages individually, we can gather more fine-grained insights into the current coverage and gaps of the tools landscape. Similarly to the above categorisation of the 216 tools, we determined whether a tool supported a stage by whether it provided *actionable* support, i.e. the mere mentioning or recommending of a specific action was not sufficient. After careful consideration, we decided to omit the first stage of the SHI process framework in this mapping exercise: “*Achieving a Positive Organisational Attitude Towards SHI*”. This was driven by the fact that also tools that promote SHI without actionable support would support this stage, thus also leading to an overlap with Category 2 from above.

4 Results: rAI Tools Support Provide Only Limited Support for SHI

We now present our analysis of the current support landscape for responsible SHI during AI development, i.e. point to actions and aspects that are currently well-supported as well as gaps.

4.1 Actionable Support for SHI is Scarce

We categorised the 216 tools by whether they provided explicit and actionable support for SHI during AI development. As illustrated on the left in Fig. 1, only 64 (30%) of tools encouraged SHI (green and blue slice). Of these, only 38 (18%, green slice) provided actionable support, i.e. went beyond abstract pointers and actually facilitated SHI. Importantly, 45 further tools (21%, red slice) did encourage the consideration of stakeholder needs or contexts *without explicitly requiring or promoting their involvement* (implications discussed in §5). The remaining 106 tools (49%, purple slice) did not mention SHI at all. These were often purely technical tools such as fairness metrics or libraries. Taken together, these findings suggests that **the majority of rAI tools do not provide actionable support for SHI**.

4.2 Tools Support Only Early Stages of the AI Lifecycle or Stages After Deployment

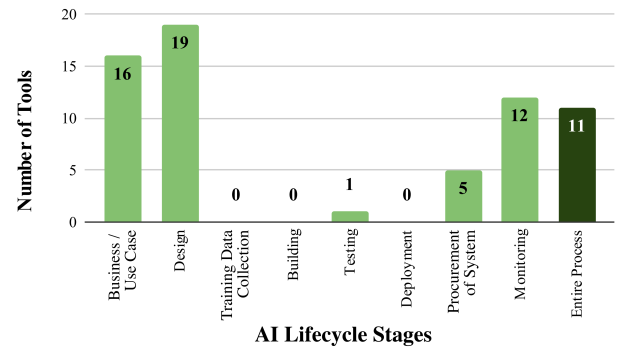
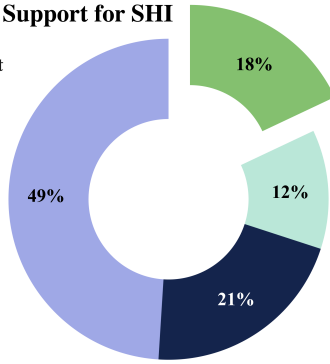
When mapping the 38 tools that did provide actionable SHI-support (i.e. from Category 1, green slice) to the AI lifecycle stages, we found that the two earliest stages of the development process were best supported (illustrated in Fig. 1, right): *Defining a Business Case/Use Case* and *Designing the System* were supported by 42% and 50% of the tools (63% of tools supported at least one of them). It is worth noting that the vast majority of these tools took AI as a given, i.e. did not explore whether AI is the most appropriate approach to achieving an objective/business case, including whether other supporting interventions would be beneficial. A second cluster of

Table 1: The seven meta-reviews from which we retrieved the 235 tools that we reviewed.

Title of Meta-Review	Published in	Number of Tools Reviewed	AI-Specific
Surveying a Landscape of Ethics Focused Design Methods [13]	2024	63	No
Guidelines for Integrating Value Sensitive Design in rAI Toolkits [59]	2024	63	Yes
Seeing Like a Toolkit: How Toolkits Envision the Work of AI Ethics [71]	2023	27	Yes
From What to How: An Initial Review of Publicly Available AI Ethics Tools, Methods and Research to Translate Principles into Practices [48]	2020	106	Yes
From Ethical AI Frameworks to Tools: A Review of Approaches [56]	2023	106	Yes
Putting AI Ethics to Work: Are the Tools Fit for Purpose? [6]	2021	25 (39 in total) ¹	Yes
No Such Thing as One-Size-Fits-All in AI Ethics Frameworks: A Comparative Case Study [57]	2023	4	Yes
TOTAL REVIEWED (minus duplicates) :		216 (+ 19 excluded)	59%

rAI Tools Categorised by Support for SHI

- Explicit & Actionable Support
- Explicit Support but Not Actionable
- Discusses Stakeholder Needs Without Demanding Actual Involvement
- No Mentioning

**Figure 1: Left: categories of rAI tools based on the extent with which they supported SHI. Right: number of tools that provided actionable support for the stages of the AI lifecycle after Ayling and Chapman [6] (tools could support multiple stages).**

tools supported SHI after a system was deployed, i.e. when *Procuring* or *Monitoring* it (13% and 29%). All other stages in the middle of the development lifecycle—i.e. *Training Data Collection*, *Model Building*, *Testing*, or *Deploying*—were not at all considered in SHI tools. The only exception was Madaio et al. [45]’s fairness checklists that guided practitioners through a socio-technical assessment of fairness that required the involvement of affected communities. Even the eleven tools that could be applied throughout the entire AI lifecycle provided no support specific to SHI during these stages [e.g. 31, 32, 42]. Overall, these findings suggest that SHI is well-supported for early ideation stages and after the system is deployed while the **intermediate AI lifecycle stages were rarely considered**. We will discuss this pattern in §5.

4.3 Gaps in Tool Support Along the SHI Process

While the previous section reported on the findings when mapping the 38 SHI-supporting tools from category 1 to the AI lifecycle, this section reports on the results of mapping these to the SHI process (§3). The process is illustrated at the bottom of Fig. 2. We identified that 63% of the tools supported the communication with the stakeholder about their needs and values during a SHI session. Interestingly, the stages after a session were far less well-supported: only four tools aided the storage of gathered insights, six their communication with colleagues, and four the prioritisation of those. The very last stage—informing stakeholders on the impact that

their input had on system design—was not supported at all. This suggests that **many other aspects of SHI are currently not well supported**, further confirmed by the fact that only one tool aimed at supporting the *Gathering of Required Resources* stage, i.e. helped to estimate costs.

The second most-supported stage of the SHI process was *Motivating the Correct Scope* of SHI (15 tools, 39%). This scope refers to three aspects: (i) all affected communities should be involved/ receive a voice in the process (especially at-risk groups), (ii) these groups have to be empowered/granted the agency to actually shape the system, and (iii) such SHI should start at the very outset of the AI lifecycle so that affected communities can impact system objectives [38]. The tools supporting this stage mainly supported the first aspect, i.e. supported SHI with a broad range of stakeholders such as through stakeholder mapping exercises. However, tool support for the actual recruitment of such broad range of stakeholders (i.e. aiding the stage *Recruiting a Representative and Inclusive Sample*) was rare: only three tools support this step despite previous research describing this as very important, especially for vulnerable groups [10, 28, 55]. Importantly, these three tools all regarded SHI after the system was deployed, i.e. supported inclusive SHI by informing the public to be aware and critical about the AI systems implemented around them/affecting their lives. While this is a very valuable form of SHI, this meant that no tool helped practitioners to access stakeholder groups during system development. Taken together,

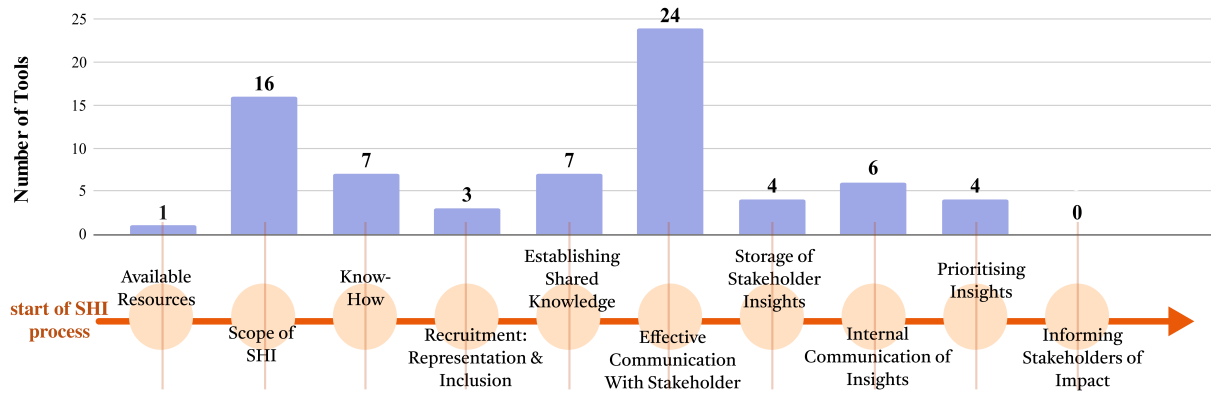


Figure 2: Number of SHI-specific rAI tools that support each of the steps of the responsible SHI process framework after Kallina and Singh [38] (tools could support multiple steps).

the identified patterns suggest that **actions required for inclusive SHI samples are not well supported by current tooling**. In the next sections, we will bring our findings together and discuss potential reasons for current gaps, implications, and concrete actions to advance better practitioner support going forward.

5 Discussion: Tool Support for Responsible SHI is Fragmented

We present an analysis of the rAI tools available to practitioners for the implementation of responsible SHI practices during AI development. Our findings emphasise that responsible SHI practices are currently not well supported by rAI tools – despite prominent calls for rAI-advancing SHI in the literature (§2.1). This analysis helps to direct practitioners towards available tools and literature, as well as to highlight gaps in practitioner support to the rAI and HCI research communities to inform future research efforts. More concretely, we found:

- Less than 20% of the reviewed rAI tools provided actionable support for responsible SHI practices, implying that the **push for SHI in the rAI literature (§2.1) is not yet reflected in tools** available for its concrete implementation.
 - Some **tools even risked disincentivising SHI** by suggesting that practitioners can identify and discuss stakeholder needs without giving affected communities any voice or agency in this process. Since practitioners will not be able to account for the needs of stakeholders that have very different lived experiences to their own, e.g. marginalised or at-risk communities who are still very rare among AI practitioners [65], their perspectives and needs are likely to be forgotten. This implies that the benefits of SHI for rAI efforts (§2.1) cannot be advanced.
- **The arguably most obvious aspect of SHI is well-supported, i.e. stakeholder communication** during SHI sessions to explore their needs, especially at early stages of system development and we hope to direct practitioners towards this available support. This concentration of tools might be caused by the often only superficial involvement of AI practitioners in scoping or creating rAI tools [40] [for rare but excellent exceptions see 24, 40, 45]. Thus, the most obvious aspect of SHI, i.e. speaking to a stakeholder, received the bulk of attention (cf less obvious but required actions, see next points). Note: most tools for early ideation stages took it as a given

that AI will be central to the solution, i.e. other approaches to achieve an objective were not considered.

- **Many actions when planning an SHI session are not well supported**, more specifically:
 - Tools that help to calculate and argue for the required resources to conduct the assessment.
 - Support in recruiting more inclusive samples, e.g. a clear, stepwise plan on how to approach marginalised communities or pointers to relevant civil society groups with established access to such groups.
- **Support for reactions to gathered insights are very scarce**, mirroring the findings of Berman et al. [9], i.e. that tools are mainly evaluated by their usability vs their impact. More specifically, we find lacking:
 - Tools facilitating the effective storage of gathered insights [for an exception see 8], such as, for example, through LLM-based systems which might further facilitate the communication of these insights to colleagues.
 - A systematic mechanism for the prioritisation of (potentially conflicting) insights, e.g. a process map or canvas/table with weights for different feedback points, thereby also incentivising reactions to gathered feedback and the traceability of decisions (increasing the stakeholder’s agency after [14, 17]).
 - Platforms open to the involved stakeholders to demonstrate the impact they had on the system.
- **SHI during pre-deployment system assessments are scarce** despite appearing to be a promising avenue to incentivise reactions to identified needs (more in §6.2).

Taken together, the currently patchy tool landscape, even including some misleading tools, is detrimental: it is a fundamental barrier to responsible SHI in the AI practice. To address these urgent concerns, we highlight key reactions in §6.

5.1 Limitations

This work was designed to scope and motivate future work, the derived recommendations for such are in §6.3. Thus, we acknowledge that our list of rAI tools is not a complete list of all available tools; rather, we aimed at capturing a broad range of prominent tools to explore their coverage, i.e. the tools that practitioners are likely to encounter as identified from key studies. Further, this work does not involve AI practitioners themselves. Thus, we might have

Table 2: Derived research questions, designed as a starting point to react to and expand on the findings of this work.

IMMEDIATE RESEARCH AGENDA FOR rAI-ALIGNED SHI	
Co-Designing Tools to Prepare & React to SHI Sessions	
-	What types of tools would help practitioners to gather the resources required for responsible SHI?
-	How can we support the recruitment of stakeholders, especially of marginalised communities requiring rapport-building?
-	How can the storage and internal communication of SHI insights be supported? Can this motivate reactions to these insights?
-	What would help to ensure that SHI insights are prioritised, even if in conflict with commercial interests?
-	How can we better support the collection and reaction to insights that suggest that AI (alone) is not the most valuable approach to achieve an objective?
Exploring and Supporting SHI During Testing & Deployment	
-	How could affected communities be included during pre-deployment tests? How can we ensure their agency thereby?
-	How could a system development approach be realised in which affected communities guard deployment, i.e. they have to consider the system as sufficiently safe, suitable, and beneficial to agree to its deployment?
-	What would be the limits of the stakeholders' decision agency thereby, e.g. can they veto but not force deployment?
Especially relevant in safety-critical or highly regulated domains where the deployment of unsafe systems would be detrimental.	
Gathering Insights Into Practitioner Perspectives & Practices	
-	What are the key bottlenecks of responsible SHI in practice? How do practitioners currently address those?
-	What are tools or practices that are not captured in the academic literature that practitioners use to support SHI efforts?
-	How do practitioners perceive the difference in conducting responsible vs commercially motivated SHI?
-	Does this affect key bottlenecks? Are there issues in conducting rAI-aligned SHI in the AI practice that cannot be addressed by tools (i.e. not <i>awareness</i> but <i>incentive</i> issues [38])?

missed tools or workarounds that practitioners deploy in practice to support their SHI efforts. However, this study analysed tools specifically created to support SHI during technology development (vs tools that practitioners repurposed to do so). By combining several of the most prominent rAI meta-reviews of the last four years, we expect to have covered broad trends of such tools. We encourage future work to involve practitioners to investigate and learn from their potential workarounds to inform more effective tool support going forward. We provide a starting point in §6.3.

6 Proposing Immediate Action for the HCI & rAI Community

This work emphasises the lack of practitioner support in implementing rAI-aligned SHI in the AI practice, and thus identifying gaps in the current research and tools landscape. This lack of support is concerning in a time in which rAI literature increasingly promotes SHI and its benefits (§2.1) as it leaves practitioners with a call-to-action but insufficient actionable guidance. We urge the HCI and rAI communities to address this gap in a timely manner, and we next outline some concrete actions as a starting point. Across all actions, we urge both communities to work together as well as in close and continuous collaboration with AI practitioners and affected communities, especially at-risk groups.

6.1 Co-Designing rAI-Aligned SHI Tools to Address Current Gaps

Our findings show the clear need to address the disconnect between increasing calls for SHI in the rAI guidance/literature and the lack of tools that operationalise this for AI practitioners. This apparent 'implementation gap' represents a clear opportunity for the research community to further advance rAI-aligned SHI tools going forward. Our insights into current coverage and gaps (§5) might serve as a starting point to scope such tools and research; §6.3 and §6.2 point towards concrete examples for future studies. In particular, we urge the development of tools that cover and support the entire SHI

process, as no such tool was identified in our analysis. Throughout these efforts, we encourage for participatory approaches with AI practitioners and rAI experts to ensure that resulting tools meet the practitioners needs whilst being aligned with rAI goals.

6.2 Exploring SHI During Pre-Deployment Assessments

Currently, SHI during the testing stage (which includes readiness-assessments) seems completely ignored by current tooling (§4.2). This reflects a wider gap in the rAI literature: SHI at these stages is rarely discussed with the exception of technical assessments through external experts (e.g. red-teaming or bounties [23, 67]) or audits that promote SHI during pre- or post-deployment audits [e.g. 43, 72]. For example, work concerning system development approaches in which the system's most affected communities determine whether a system is ready to be deployed seem to not have been considered. We see a real opportunity in such approaches to facilitate stakeholder empowerment throughout the development process, i.e. that their need and concerns actually impact the system's design as they 'gate-keep' deployment. Further, this would help in translating more abstract rAI aspects (e.g. *transparency*, *privacy*) to a specific application context, as the affected communities would define themselves when this aspect is met.

6.3 Immediate Research Agenda to Explore & Support Responsible SHI Practices

To address the identified disconnect now, Table 2 outlines an immediate research agenda as a starting point to address current gaps and areas of uncertainty. In particular, we call for more insights into whether tools alone can achieve rAI-aligned SHI in practice, and how effective they are. Tools can provide knowledge and awareness [38]; however, our finding that support for reactions to stakeholder insights are scarce might hint at the fact that power-asymmetries might hinder this step – something that could not be addressed by tools alone. Thus, we require further insights into current barriers

of/bottlenecks to rAI-aligned SHI in the AI development practice, some first questions are included below (or an exploration of some challenges, see [38]).

7 Conclusion

This work emphasises the implementation gap in rAI-aligned SHI: despite growing calls for SHI that advances rAI efforts, tools that support this in practice appear very limited and focused on some actions only while many aspects are not at all supported. It is essential to highlight this disconnect so that the community can react in a timely and informed manner in order to match the increasing calls for rAI-oriented SHI with actionable tools that support its implementation. Going forward, we hope to motivate *three key research efforts*: (i) participatory studies for better, rAI-aligned tools that support the entire SHI lifecycle, (ii) explorations of how SHI during pre-deployment assessments could be realised, and (iii) what else besides tool support might be required to advance rAI-aligned SHI in practice. In all, we hope to set the stage for future works that contribute to more rAI development practices and therefore more responsible systems.

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